

ML Lab

From interactive exhibits to a complete machine learning curriculum

A strategic review, curriculum architecture, release roadmap, and product recommendations for maximum educational impact.

● SEE

● RUN

● BREAK

● EXPLAIN

NAVIGATION

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STRATEGIC THESIS

ML Lab should not become an encyclopedia of algorithms. Its strongest position is to become the best interactive curriculum for understanding how machine learning systems learn, fail, mislead, and survive contact with the real world.

EXECUTIVE SUMMARY

A distinctive pedagogy with room to become a complete learning system

ML Lab already has a compelling core: learners do not merely read about a concept; they manipulate it, observe what changes, deliberately create a failure, and explain why it happened. The site's **See - Run - Break - Explain** pattern is the product's clearest differentiator.

The current curriculum forms a coherent introduction to supervised learning, especially linear models, optimisation, generalisation, leakage, and regularisation. The next step is not simply to add more algorithms. It is to broaden the learning journey around data, evaluation, uncertainty, deployment, modern generative AI, and the human consequences of automated decisions.

01

Keep the pedagogy

Preserve the experiment-first sequence. It builds intuition, implementation awareness, failure diagnosis, and transfer.

02

Fill the practical gap

Prioritise data quality, validation, metrics, calibration, subgroup performance, and distribution shift.

03

Expand by assumptions

Teach model families through the assumptions they make and the characteristic ways they fail.

04

Build toward systems

Connect model learning to deployment, monitoring, safety, and human oversight - not as optional extras.

HIGHEST-IMPACT FIRST RELEASE

Launch an **Honest ML** journey: baselines; train, validation, and test sets; sampling bias; missing and categorical data; class imbalance; metric selection; calibration; subgroup evaluation; distribution shift; and a cumulative diagnosis capstone.

Active-learning evidence supports this overall direction: learners generally perform better when they actively work with ideas rather than receive them only through conventional exposition.^[2]

CURRENT PRODUCT REVIEW

What is working - and what currently limits the curriculum

What is already working

The pedagogy is the product. The laboratory sequence is more powerful than adding decorative interactivity to a lecture. Learners form a prediction, act, inspect a consequence, and articulate a causal explanation.

Failure modes appear early. Leakage, overfitting, unstable test splits, bias-variance, and regularisation are treated as central learning objects rather than end-of-course caveats.

The dependency language is useful. 'Builds on', 'leads to', and 'compare with' encourage exploration without removing the sense of progression.

The writing creates memorable mental models. Short phrases such as 'the loss is the judge' translate well into visual experiments.

What limits the next stage

The curriculum is still predominantly a **linear supervised-learning spine**. It stops before most real data problems, alternative model families, unsupervised learning, uncertainty, deployment, modern generative AI, and responsible system design.

- Distinguish class labels, decision scores, and calibrated probabilities.
- Resolve the XOR wording inconsistency: a linear boundary can classify three of four canonical points, but cannot solve all four.
- Separate validation data used for model choice from the final test set used once for unbiased evaluation.
- Make the Gradient an explicit mathematical bridge between loss and gradient descent.
- Add cumulative proof of learning through diagnosis and capstone challenges.

POSITIONING OPPORTUNITY

The strongest market and learning position is not 'learn every ML algorithm'. It is 'learn how intelligent systems behave under pressure' - with failure, uncertainty, values, and deployment built into the experience.

Broader machine learning curricula commonly extend beyond supervised foundations into data preparation, embeddings, unsupervised learning, production systems, and fairness.^[3]

STRATEGIC DIRECTION

A north star for the expanded curriculum

A learner should leave ML Lab able to inspect a dataset, establish a baseline, choose and evaluate a model, expose its failure modes, communicate uncertainty, deploy it cautiously, and identify who could be harmed when its assumptions stop holding.

Six design principles

- 01 Experiment before exposition**
Begin with a prediction and an observable system, then introduce formal language once the learner has something concrete to explain.
- 02 Failure by design**
Every major concept should include a deliberate way to break it, plus guidance on how the failure would appear in a real project.
- 03 Teach assumptions, not catalogues**
Model families become memorable when learners see what each assumes and what evidence contradicts that assumption.
- 04 Make decisions explicit**
Metrics, thresholds, uncertainty, and error costs should be connected to the decision a model supports.
- 05 Treat the model as one component**
Deployment, monitoring, feedback loops, and human escalation belong in the core curriculum.
- 06 Attach an impact lens**
Each failure should ask who experiences it, whether aggregate metrics reveal it, and how the decision can be contested.

CURRICULUM ARCHITECTURE

Nine connected journeys - one coherent learning system

The curriculum should expand as a sequence of connected journeys. Each journey answers a distinct learner question while reinforcing the same laboratory method.

01 Foundations 2.0 Build honest first models with baselines, valid splits, and causal humility.	02 Data and features Understand how representation, missingness, sampling, and pipelines shape results.	03 Evaluation and decisions Choose metrics, express uncertainty, calibrate confidence, and diagnose shift.
04 Model workshop Compare families through assumptions: neighbours, trees, forests, boosting, margins.	05 Discovering structure Learn without labels through clustering, components, anomalies, and embeddings.	06 Deep learning Move from computation graphs to representations, attention, and transformers.
07 Generative AI literacy Understand tokens, generation, retrieval, evaluation, tool use, and attacks.	08 From model to system Frame, deploy, test, monitor, roll back, and improve ML in production.	09 Responsible AI Connect fairness, privacy, interpretability, security, oversight, and accountability.

CROSS-CUTTING LENSES

Uncertainty, fairness, safety, interpretability, communication, and human oversight should recur across all journeys rather than appear only in a final ethics module.

JOURNEY 01

Foundations 2.0: make the first model an honest model

Core additions

- Baselines: mean, median, majority class, and simple decision rules.
- Train, validation, and test sets - with distinct purposes.
- Prediction versus causation.
- The Gradient as an explicit mathematical detour between loss and gradient descent.
- A final challenge that combines task choice, splitting, fitting, metrics, leakage prevention, and explanation.

CAPSTONE CONCEPT

Give learners a mystery dataset and ask them to choose the task, establish a baseline, create a valid split, train a model, choose a metric, and explain one consequential failure.

Why this matters

The current foundations are technically coherent, but they can give the impression that fitting a model is the centre of the work. In practice, honest comparison, valid evaluation, and disciplined problem framing determine whether the result means anything.

A redesigned foundation should make **epistemic discipline** visible from the beginning: What is the baseline? What information was available at prediction time? Which data informed choices? What remains genuinely unseen?

This also creates the vocabulary needed for every later journey - especially model selection, uncertainty, production monitoring, and responsible AI.

DESIGN RULE

No learner should be allowed to celebrate an accuracy score before seeing the baseline, the split, the metric, and the error distribution.

Journey 02 preview: data and features

This should be the first major content expansion. It offers more practical value than immediately adding a long list of algorithms because real-world success often depends on who is represented, how variables are encoded, and whether the pipeline preserves valid evaluation.

JOURNEY 02

Data and features: make the hidden modelling choices visible

Exhibit	Central experiment
Sampling and selection bias	Change who enters the dataset and watch an apparently accurate model fail on the population it is meant to serve.
Missing data	Switch between missingness mechanisms and compare dropping rows, simple imputation, and missingness indicators.
Categories are not numbers	Encode unordered categories as integers, then repair the artificial ordering with one-hot or learned representations.
Outliers, noise, and bad labels	Separate measurement error, label error, and legitimate rare cases; observe how each changes the model.
Feature transformations	Turn a curved relationship into one a linear model can learn - and show when transformations distort meaning.
Feature interactions	Reveal how two individually weak features become predictive when they are considered together.
Class imbalance	Build a 99%-accurate classifier that finds none of the important cases, then change the metric and threshold.
Data pipelines	Move preprocessing outside cross-validation to create leakage, then repair the workflow with a fitted pipeline.

WHERE THE IMPACT IS

This journey turns ML Lab from a model-demonstration site into a realistic problem-solving environment. It equips learners to notice the subtle data decisions that dominate many failed projects.

JOURNEY 03

Evaluation, uncertainty, and decisions: the highest-impact expansion

This journey is the natural extension of ML Lab's existing strength in failure modes. It connects predictive performance to the decisions a system supports - and to the people who experience its errors.

Exhibit	Central experiment
The baseline to beat	Compare the model with the simplest credible alternative before interpreting improvement.
Validation is not testing	Tune repeatedly on a 'test' set until its score becomes misleading, then reserve a true final holdout.
Metrics encode values	Change the cost of false positives and false negatives and watch the preferred model and threshold change.
ROC or precision-recall?	Alter prevalence and see why a strong ROC curve can coexist with poor precision in rare-event problems.
Calibration	Group predictions by stated confidence and compare '80% confident' with what actually happens 80% of the time.
Prediction intervals	Move from point predictions to ranges and make coverage and sharpness visible.
Subgroup evaluation	Preserve overall accuracy while degrading performance for one population - then reveal the hidden disparity.
Distribution shift	Move the deployment population away from the training distribution and observe silent performance decay.
Error analysis	Turn recurring errors into a concrete data-collection, feature, threshold, or modelling intervention.

EDUCATIONAL AND SOCIETAL LEVERAGE

Learners who understand calibration, subgroup performance, error costs, and shift are better prepared to build useful systems and less likely to mistake a single aggregate score for evidence of reliability.

NIST's AI risk framework similarly treats validity, reliability, safety, security, transparency, interpretability, privacy, and fairness as interconnected properties of an AI system.^[4]

JOURNEYS 04-05

Teach models by the assumptions they make

Rather than assembling a model zoo, use each family to expose a different way of representing evidence - and a characteristic way that representation fails.

Journey 04 - The model workshop

Nearest neighbours. Learning by similarity; sensitivity to scale, noise, k , and dimensionality.

Decision trees. Learning by questions; purity, depth, instability, and interpretability.

Random forests. Reducing variance through bootstrapping and feature diversity.

Gradient boosting. Sequentially repairing residual errors with small trees.

Support-vector machines. Margins, support points, and kernels.

Naive Bayes. Combining evidence while deliberately breaking independence.

Hyperparameter search. Exposing winner's curse after trying many configurations.

Choosing a model. Balancing accuracy, interpretability, latency, robustness, and data needs.

Journey 05 - Discovering structure without labels

Distance and similarity. Make geometry and scaling the first-class concepts.

K-means. Show failure on non-spherical, unequal-density, and badly scaled clusters.

Hierarchical clustering. Explore how linkage choices change the story.

Density-based clustering. Discover arbitrary shapes and treat noise explicitly.

Principal component analysis. Compress variance while revealing what information disappears.

t-SNE and UMAP. Demonstrate how compelling visual layouts can mislead.

Anomaly detection. Separate rarity from meaningful abnormality.

Recommendation and embeddings. Connect learned geometry to retrieval and modern AI.

SUGGESTED SEQUENCING

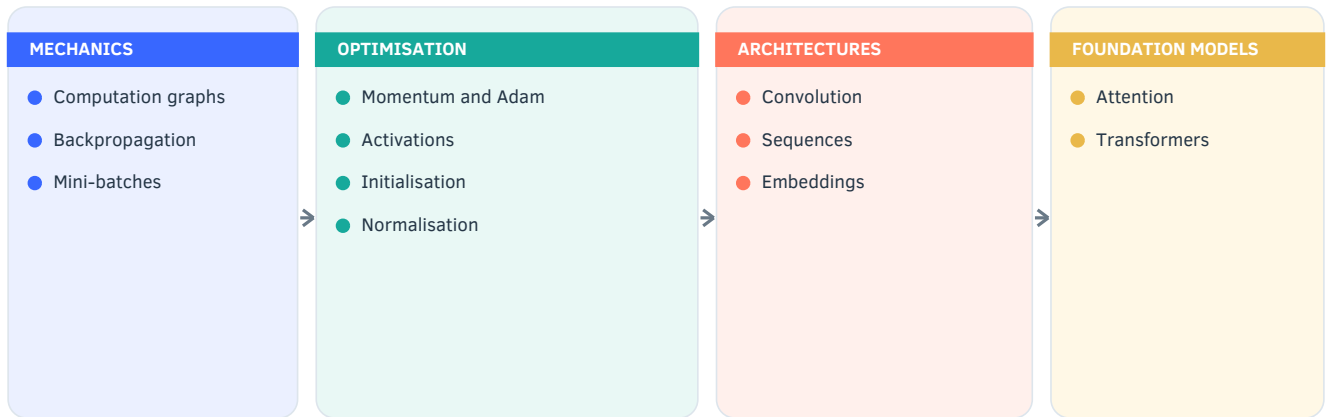
Trees, forests, boosting, nearest neighbours, clustering, and PCA should come first. They broaden the learner's mental model quickly, remain highly visual, and expose assumptions that differ sharply from linear models.

A broad machine learning foundation normally includes clustering and dimensionality reduction alongside supervised learning.^[5]

JOURNEY 06

Deep learning and representations: extend XOR into a mechanics-first journey

The current neural-network exhibit provides a natural launch point. The next journey should reveal how networks compute, learn, become unstable, and build representations - before introducing modern architectures.



One consequential knob per exhibit

ACTIVATIONS Let learners kill a network by pushing units into saturation or by creating dead ReLUs.	INITIALISATION Start too small or too large and watch signals or gradients vanish and explode.
OPTIMISERS Keep the loss surface fixed and compare plain SGD, momentum, and Adam trajectories.	NORMALISATION Change batch composition and show how internal scales alter training dynamics.

The objective is not to display every architecture. It is to let learners predict and diagnose the mechanics that make deeper systems trainable.

JOURNEY 07

Generative AI and LLM literacy: audience growth without losing rigour

This journey can produce the strongest acquisition and sharing impact, but it should be built on probability, embeddings, optimisation, and evaluation rather than presented as a disconnected prompt-engineering course.

Mechanics

- Tokens are not words
- Next-token prediction
- Temperature, top-k, and top-p
- Context windows
- Attention

Adaptation and retrieval

- Pretraining versus instruction tuning
- Embeddings and semantic search
- Retrieval-augmented generation
- Prompting versus fine-tuning versus retrieval

Evaluation and safety

- Hallucination and grounding
- LLM evaluation
- Prompt injection and untrusted content
- Quality, latency, and cost trade-offs

Systems and agency

- Tool use
- Structured outputs
- Agents and control loops
- Human approval and failure recovery

CURRICULUM RULE

Do not make prompting the foundation. Make generation mechanics, grounding, evaluation, security, and control the foundation - then show prompting as one interface to the system.

Contemporary introductory ML curricula increasingly place embeddings and LLMs alongside production systems and fairness, connecting modern AI to core machine learning principles.^[3]

JOURNEY 08

From model to system: the greatest career and applied-project impact

A high-performing notebook model is only one stage in a dependable product. This journey should turn the entire system lifecycle into an observable experiment.



Core exhibits

- Frame the problem and define success.
- Establish a non-ML baseline.
- Version data, code, and models.
- Compare offline and online evaluation.
- Create and repair train-serve skew.
- Choose batch or real-time inference.
- Balance latency, throughput, and cost.

Reliability exhibits

- Test the ML pipeline, not only the model function.
- Use shadow deployment and staged rollout.
- Run an A/B test and interpret uncertainty.
- Monitor performance, drift, and data quality.
- Expose feedback loops and delayed labels.
- Design rollback and human escalation.
- Plan continual learning without silent regressions.

WHY THIS JOURNEY MATTERS

It connects conceptual learning to employable practice. Learners see that production success depends on problem definition, interfaces, monitoring, experiments, failure recovery, and user experience - not only model training.

Full Stack Deep Learning makes the same central point: building an AI product extends beyond training into problem definition, deployment, monitoring, continual learning, and user experience.^[6]

JOURNEY 09

Responsible and human-centred AI: a journey and a layer

Responsible AI should have dedicated exhibits, but it should also change the questions asked in every other journey. A system can be statistically strong and still be unsafe, unfair, unaccountable, or impossible to contest.

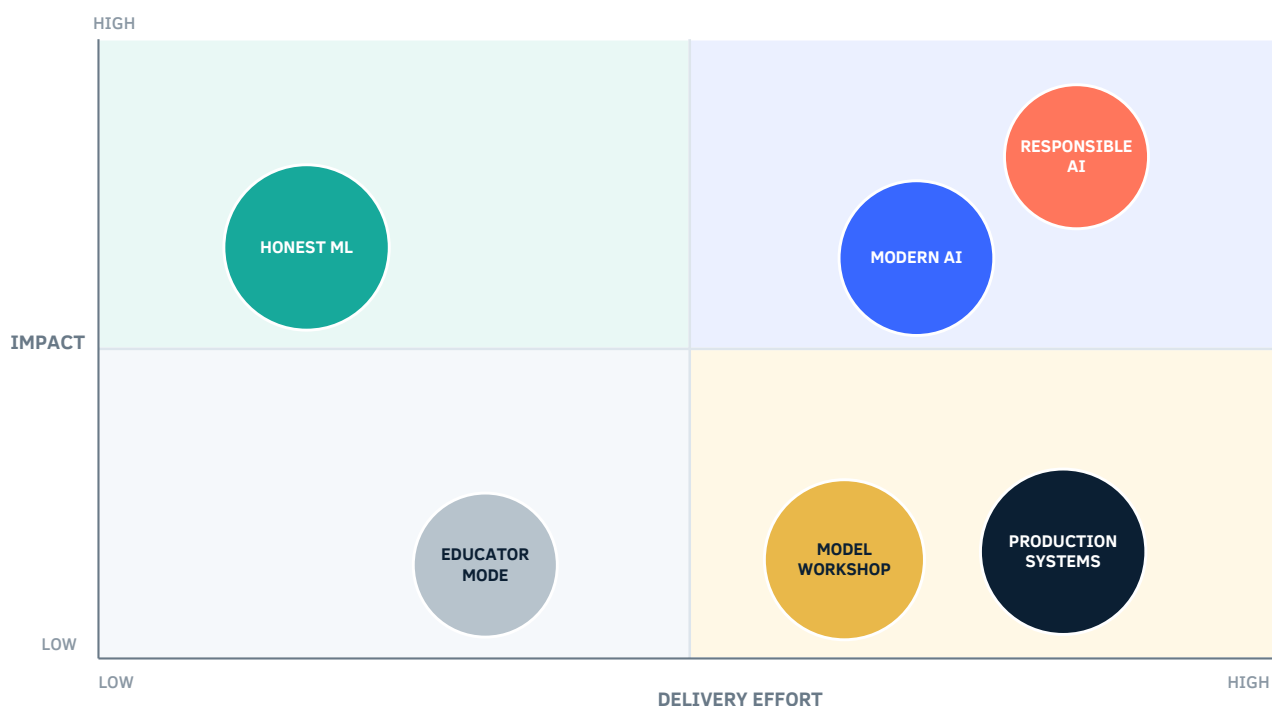
Exhibit	Central experiment
Fairness definitions	Compare group metrics and reveal trade-offs that cannot always be satisfied simultaneously.
Global and local interpretability	Show how explanations answer different questions and can disagree.
Proxy variables	Remove a protected attribute while preserving information about it through correlated features.
Privacy and memorisation	Measure when a model reproduces sensitive or rare training information.
Membership inference	Infer whether a record was in training from the model's confidence behaviour.
Differential privacy	Trade utility for bounded information leakage and make the privacy budget visible.
Poisoning and adversarial examples	Alter a small amount of data or input structure to trigger large changes.
Human oversight	Expose automation bias, escalation design, contestability, and accountability.

THE IMPACT LENS
Who experiences this failure?
Would the aggregate metric reveal it?
Who can contest or override the decision?
What monitoring would detect the harm early?

PRIORITISATION

Where the most impact can be had

The expansion should be sequenced by educational leverage, product differentiation, implementation feasibility, and audience demand. The matrix below is directional: it highlights where a relatively focused release can change the value of the whole platform.



Interpretation

Honest ML	Highest near-term leverage. It deepens the current proposition, reuses existing interaction patterns, and prevents the most consequential beginner mistakes.
Modern AI	Highest audience-growth potential. It should follow embeddings, probability, and evaluation so it strengthens rather than dilutes the curriculum.
Responsible AI	High impact across the roadmap. Build the impact lens early, then expand dedicated exhibits as the system curriculum matures.
Production systems	High career impact and product maturity, but more complex to simulate well. Sequence after evaluation and model breadth.

RELEASE ROADMAP

A four-release sequence that compounds learning and product value

01 Honest ML

Baselines; train/validation/test; sampling bias; missing and categorical data; imbalance; metric costs; calibration; subgroup evaluation; shift; diagnosis capstone.

02 Models that think differently

Decision trees; random forests; boosting; nearest neighbours; model choice; clustering; PCA; anomaly detection. Teach assumptions before breadth.

03 How modern AI works

Backpropagation; embeddings; attention; transformers; token generation; retrieval; hallucination; LLM evaluation; prompt injection; tool use.

04 From demo to dependable system

Deployment; offline/online experiments; monitoring; drift; feedback loops; privacy; fairness; security; human oversight; rollback.

DEFER UNTIL THE CORE IS STRONGER

Reinforcement learning, GANs, specialised vision architectures, and advanced Bayesian methods are visually attractive, but they offer less immediate benefit than data quality, evaluation, model families, generative AI literacy, and production reliability.

What makes the sequence compound

- Release 01 provides the evaluation language required to compare every later model and system.
- Release 02 broadens mental models while preserving highly visual, interactive pedagogy.
- Release 03 captures modern interest but rests on real foundations rather than interface tricks.
- Release 04 turns conceptual knowledge into dependable, responsible practice.

PRODUCT DESIGN

Product changes that amplify the curriculum

01

Cumulative lab notebook

Record each learner's prediction, manipulation, observation, and explanation. Generate a shareable lab report rather than only marking an exhibit complete.

02

Mystery challenges

Do not reveal whether the problem is leakage, imbalance, miscalibration, overfitting, or shift. Diagnosis is a stronger test than following a labelled demonstration.

03

Story mode and Lab mode

Use a mobile-friendly guided narrative with predetermined experiments, while preserving the full desktop laboratory for free manipulation.

04

What this exhibit simplifies

Add a concise disclosure that protects learners from turning a useful metaphor or toy dataset into a misconception.

05

Educator mode

Support shareable exhibit states, challenge presets, discussion prompts, and anonymous classroom results so the site becomes a teaching instrument.

Measure learning transfer, not only completion

A

Diagnostic transfer

Can learners identify an unseen failure without being told its label?

B

Explanation quality

Can learners connect a visual change to mechanism, evidence, and consequence?

C

Applied confidence

Can learners frame, evaluate, and communicate a small end-to-end project?

FINAL RECOMMENDATION

Build the world's most useful laboratory for machine learning judgement

ML Lab's advantage is not the number of topics it can cover. It is the way it can make invisible assumptions, failures, and trade-offs tangible. The broader curriculum should protect that advantage by expanding outward from honest evaluation and diagnosis - not by adding disconnected content.

The strongest version of ML Lab teaches learners not just how to make a model work, but how to know when it should be trusted, how it will fail, and what to do next.

References

- [1] ML Lab. Public website and exhibit catalogue. ml-lab-xi.vercel.app
- [2] Freeman, S. et al. (2014). Active learning increases student performance in science, engineering, and mathematics. *Proceedings of the National Academy of Sciences*. [Source](#)
- [3] Google for Developers. Machine Learning Crash Course curriculum. [Source](#)
- [4] National Institute of Standards and Technology. AI Risk Management Framework resources and trustworthiness characteristics. [Source](#)
- [5] Stanford University. CS229: Machine Learning. [Source](#)
- [6] Full Stack Deep Learning. Course materials on the full lifecycle of AI products. [Source](#)

Report prepared as a curriculum strategy conversion of the preceding website review. Links are included for source context; the recommendations are strategic judgements based on the reviewed experience.